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V^2-SAM: Marrying SAM2 with Multi-Prompt Experts for Cross-View Object Correspondence

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Abstract

Cross-view object correspondence, exemplified by the representative task of ego–exo object correspondence, aims to establish consistent associations of the same object across different viewpoints (e.g., ego-centric and exo-centric). This task poses significant challenges due to drastic viewpoint and appearance variations, making existing segmentation models, such as SAM2, non-trivial to apply directly. To address this, we present V^2-SAM, a unified cross-view object correspondence framework that adapts SAM2 from single-view segmentation to cross-view correspondence through two complementary prompt generators. Specifically, the Cross-View Anchor Prompt Generator (V^2-Anchor), built upon DINOv3 features, establishes geometry-aware correspondences and, for the first time, unlocks coordinate-based prompting for SAM2 in cross-view scenarios, while the Cross-View Visual Prompt Generator (V^2-Visual) enhances appearance-guided cues via a novel visual prompt matcher that aligns ego–exo representations from both feature and structural perspectives. To effectively exploit the strengths of both prompts, we further adopt a multi-expert design and introduce a Post-hoc Cyclic Consistency Selector (PCCS) that adaptively selects the most reliable expert based on cyclic consistency. Extensive experiments validate the effectiveness of V^2-SAM, achieving new state-of-the-art performance on Ego-Exo4D (Ego–Exo object correspondence), DAVIS-2017 (video object tracking), and HANDAL-X (robotic-ready cross-view correspondence). Codes and Models will be released at V^2-SAM.

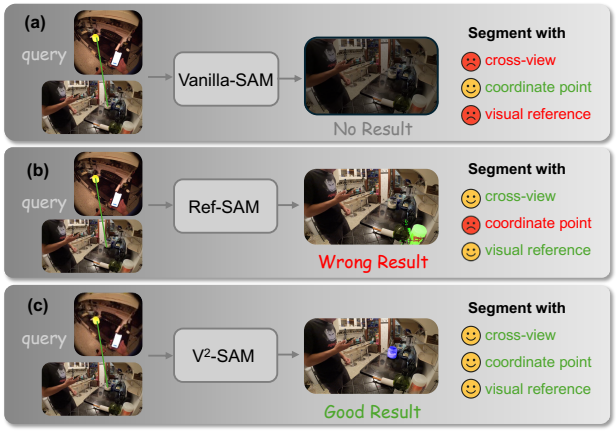


Figure 1. Comparison of SAM variants in segmentation capability. Our proposed V²-SAM supports coordinate-point and visual-reference prompts for cross-view segmentation.

1. Introduction

Cross-view object correspondence [10] aims to associate object identities across viewpoints by segmenting the same object(s) in a target view, given their mask annotation in a query view. This ability is fundamental for multi-view scene understanding [17, 24, 29, 46], video perception [8, 9, 28, 34, 35, 49, 51], and embodied AI [7, 20, 21, 39]. Among various cross-view scenarios, ego–exo object correspondence [12, 16, 22, 31] stands out as a particularly representative and challenging case. In this setting, the same object is simultaneously observed from a moving first-person (ego-centric) camera and a static third-person (exo-centric) camera, leading to drastic viewpoint and appearance discrepancies, complex background clutter, dynamic motions, and occlusions.

Recent advances in single-view segmentation have been driven by powerful foundation models such as SAM2 [37] (as Fig. 1(a)), which exhibit remarkable mask generation

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capability given well-localized prompts. However, extending such models to the cross-view setting is highly non-trivial. SAM2 depends on spatially grounded prompts (e.g., mask coordinates or bounding boxes) to condition its decoder, but in cross-view scenarios, the target object's position can vary drastically across views, making direct position-based prompting infeasible. While referring-based extensions [2, 37] (as Fig. 1(b) Ref-SAM) replace explicit positional prompts with visually guided prompt generators, they still face two major limitations: 1) they often fail under severe appearance changes across views; and 2) removing spatial prompts undermines SAM2's inherent strength in localization. The above-mentioned challenges motivate us to explore two central questions: ① Can we unlock SAM2's spatial prompting capability in cross-view scenarios? ② If so, can spatial and visual prompts complement each other to further enhance cross-view segmentation performance?

To address ①, we propose a novel *Cross-View Anchor Prompt Generator (V^2-Anchor)* that restores SAM2's ability to localize objects across views. Specifically, we leverage DINOv3's geometry-aware feature space to identify corresponding object regions between ego- and exo-centric images. This is achieved through cross-view feature matching followed by point stratification to establish reliable correspondences. To suppress noisy correspondences and avoid confusion with nearby distractors in cluttered scenes, outliers are removed and a centroid point is selected from the remaining matches, yielding a high-confidence coordinate in the target view. In this way, we effectively *unlock coordinate-based prompting* for SAM2 in cross-view segmentation, particularly under the ego–exo setting. As for ②, inspired by prior visual-referring segmentation models [2, 18, 41], we develop the *Cross-View Visual Prompt Generator (V^2-Visual)*, which introduces a novel *Visual Prompt Matcher* to align cross-view object representations from both feature and structural perspectives. Empirically, we observe that V^2-Anchor excels at determining *where it is*, while V^2-Visual is more effective in identifying *what it looks like*. Motivated by their complementary strengths, we design a multi-prompt expert framework that exploits both prompt types. Inspired by the Mixture-of-Experts [15] (MoE) paradigm [48, 54], we define each expert as a combination of a prompt generator and a SAM2-based mask decoder, and train three experts: one specialized in spatial prompts (Anchor Expert), one in visual prompts (Visual Expert), and one in fused spatial–visual prompts (Fusion Expert). Finally, we introduce a *Post-hoc Cyclic Consistency Selector (PCCS)* that adaptively selects the optimal expert for each object based on cross-view cyclic consistency of the predicted masks.

Formally, by integrating our V^2-Anchor, V^2-Visual, multi-expert training, and the PCCS into the vanilla SAM2, we construct **V^2-SAM** (as Fig. 1(c)), a unified cross-view object segmentation framework that effectively mitigates the inherent challenges introduced by drastic viewpoint and appearance variations discussed earlier. We conduct extensive experiments across a range of cross-view correspondence benchmarks, including Ego-Exo4D (ego–exo object correspondence), DAVIS-2017 (video object tracking), and HANDAL-X (robotic-ready cross-view correspondence). Results show that V^2-SAM achieves state-of-the-art (SOTA) performance on all three benchmarks, consistently outperforming strong baselines and existing competitors. Our main contributions are summarized as follows:

- **Unified framework:** We propose V^2-SAM, the first unified framework to architecturally adapt SAM2 to cross-view object correspondence, especially for the challenging ego–exo task.
- **Cross-view prompt generators:** V^2-Anchor, for the first time, enables coordinate-based prompting for SAM2 in cross-view scenarios; V^2-Visual enhances appearance-guided cues via a novel visual prompt matcher.
- **Expert integration and selection:** A multi-prompt expert framework with the novel PCCS module adaptively selects the most reliable expert for each instance.
- **Extensive validation:** The effectiveness of V^2-SAM is validated on Ego-Exo4D, DAVIS-2017, and HANDAL-X, achieving new SOTA performance.

2. Related Works

Cross-View Object Correspondence. Cross-view object correspondence, a key problem in ego–exo perception [25, 42], involves taking object queries from one view and predicting the corresponding object masks in another. Existing methods can be broadly grouped into two paradigms. *Matching-based methods* formulate the task as post-hoc matching. Baade et al. [1] first introduced Predictive Cycle Consistency to construct matched object pairs across views for training. More recently, O-MaMa [31] and DOMR [22] achieve cross-view segmentation by matching mask proposals from an external segmentation model, thereby limiting their learning and generalization capacity. *Learnable model-based methods*, in contrast, directly learn cross-view object segmentation. Fu et al. [10, 11] proposed ObjectRelator, an end-to-end framework for cross-view segmentation using visual and textual condition. To improve robustness and generalization, our method follows the learnable model-based paradigm. Unlike ObjectRelator, we introduce coordinate-based cues to unlock SAM2 for cross-view correspondence, a largely unexplored direction. Moreover, cross-view correspondence varies significantly across scene conditions. We further develop a multi-expert framework to effectively exploit the complementary strengths of both prompts.

Visual Reference-Guided Segmentation. Recent studies have extended the prompt-driven paradigm of the Seg-

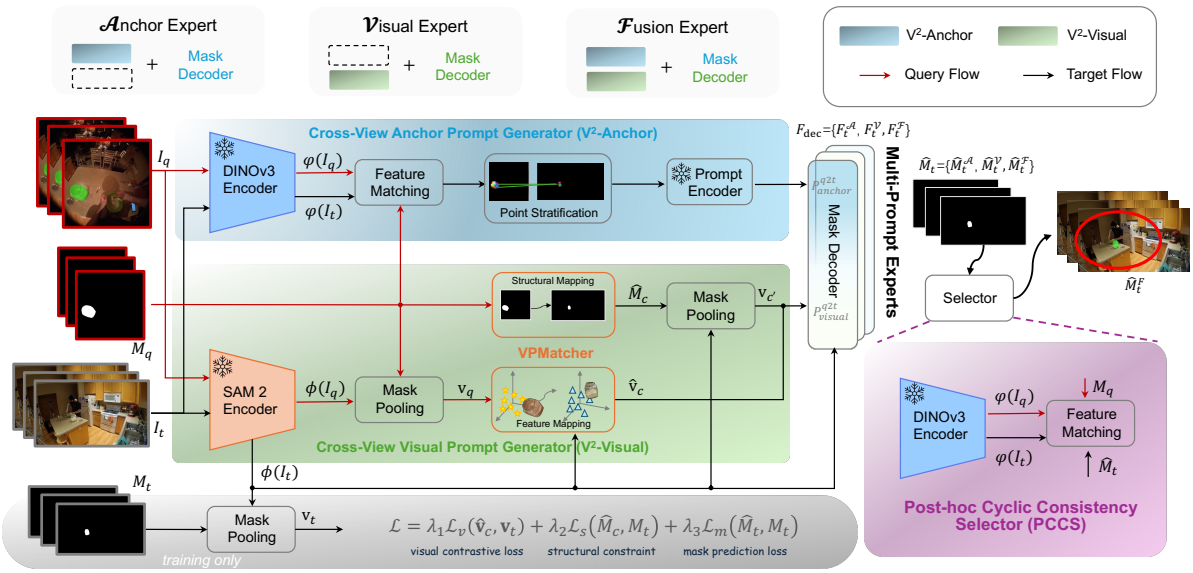


Figure 2. Overview of V^2-SAM. Given a query–target image pair (I_q, I_t) and the query object mask M_q, we generate two cross-view prompts: a geometry-guided cue P^{q2t}_{anchor} via V^2-Anchor (blue) and an appearance-guided cue P^{q2t}_{visual} via V^2-Visual (green). Each prompt is fed into its corresponding mask decoder to produce candidate target masks, and the Post-hoc Cyclic Consistency Selector (PCCS) adaptively selects the most reliable prediction as the final output.

ment Anything Model (SAM) [19] toward visual reference-guided segmentatio [30], where visual-reference images serve as prompts for segmentation. VRP-SAM [41] and ViRefSAM [2] utilize reference-image encoders to guide SAM for class-specific or domain-adaptive segmentation. Meanwhile, some studies have attempted to combine visual and textual cues for joint prompt segmentation, e.g., VLP-SAM [38], which combines visual and textual cues to improve few-shot generalization. Unlike these methods, which mainly focus on intra-view visual prompting, our work explores cross-view correspondence, emphasizing the integration of spatial and appearance cues to enable consistent segmentation. This perspective extends visual reference-guided segmentation toward more general and embodied perception scenarios.

3. Methodology

Task Formulation. Formally, given two temporally aligned image/video from distinct views, a query view I_q/V_q and a target view I_t/V_t, and a query object mask(s) M_q in the query view, the objective is to predict the corresponding object mask $\hat{M}_t$ in the target view. Ego–Exo object correspondence serves as a representative case of this formulation, in which the two views come from egocentric and exocentric cameras, respectively. Depending on the transfer direction, the task can be defined as an Ego2Exo or an Exo2Ego subtask. In all cases, the model relies solely on visual information without access to camera poses, semantic labels, or explicit 3D geometry.

3.1. Overview of Proposed V^2-SAM

SAM2 [37] is one of the most powerful segmentation models to date, demonstrating outstanding generalization ability through visual prompting. Our framework is built upon SAM2, retaining its core components. Specifically, we keep the *SAM2 Encoder* $\phi(\cdot)$, *Prompt Encoder*, and *Mask Decoder*, while discarding memory-related modules to focus on frame-level correspondence. This design allows the framework to generalize seamlessly across both cross-view image and video tasks.

The overall workflow of V^2-SAM is illustrated in Fig. 2. Given a query–target image pair (I_q, I_t) and the query object mask M_q, the model first employs our two prompt generators, i.e., *Cross-View Anchor Prompt Generator* (V^2-Anchor) and *Cross-View Visual Prompt Generator* (V^2-Visual), to transform the object information from the query view to the target view. The resulting cross-view prompts, denoted as P^{q2t}_{anchor} and P^{q2t}_{visual}, encode geometry- and appearance-guided cues, respectively. Breiefly, the V^2-Anchor branch operates on DINOv3 features to establish geometric correspondences between (I_q, I_t). It identifies reliable point-level matches, refines them via stratified filtering, and converts them into coordinate-based prompts P^{q2t}_{anchor} that localize the object in the target view. In parallel, the *V^2-Visual* branch extracts region-level representations by applying mask pooling on SAM2 features within the query mask M_q. These representations are then transferred to the target view through a learnable Visual Prompt Matcher (VPMatcher), producing the appearance-guided

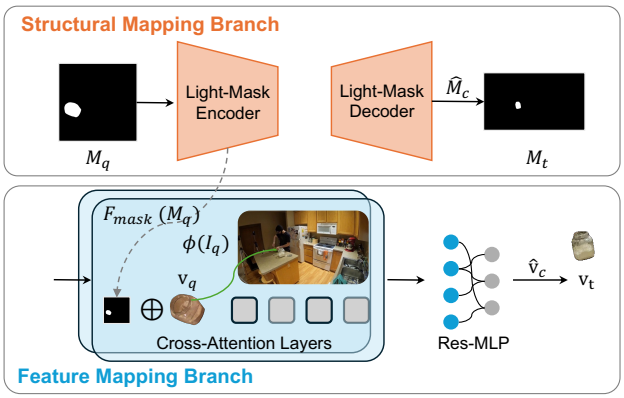


Figure 3. The structure of Visual Prompt Matcher. The Structural Mapping Branch is built upon a lightweight CNN-based mask encoder and decoder. The Feature Mapping Branch leverages Transformer-based cross-attention layers, while the Res-MLP component serves as a residual multi-layer perceptron.

prompt P^{q2t}_{visual}. With these two complementary prompts, three experts are constructed and trained: the *Anchor Expert*, the *Visual Expert*, and the *Fusion Expert*, each formed by coupling a specific prompt with the mask decoder. All experts share the same decoder architecture but maintain distinct parameters. During inference, all experts predict target-view masks, and the *Post-hoc Cyclic Consistency Selector (PCCS)* adaptively selects the most reliable output in a non-parametric manner by evaluating the cross-view mask consistency.

3.2. Cross-View Anchor Prompt Generator

The V^2-Anchor module is primarily responsible for locating the corresponding object(s) in the target view I_t by exploring geometric relationships between the query view I_q and target view I_t. To achieve this, we leverage DINOv3, which encodes fine-grained, spatially aware representations at the patch level. Specifically, we extract patch-level features $\varphi(I_q) \in \mathbb{R}^{F\times H_q\times W_q}$ and $\varphi(I_t) \in \mathbb{R}^{F\times H_t\times W_t}$, and then perform a sequence of operations, including *feature matching*, *point stratification*, *coordinate transformation*, and *prompt encoding*, to generate geometry-based anchor prompts for cross-view segmentation.

Feature Matching. For each query patch in $\varphi(I_q)$, we compute its similarity to all target patches $\varphi(I_t)$ to obtain a dense correspondence heatmap $\mathbf{H}$:

$$\mathbf{H}_{ij} = \frac{\varphi(I_q)_i^\top \, \varphi(I_t)_j}{\|\varphi(I_q)_i\|_2 \, \|\varphi(I_t)_j\|_2}, \tag{1}$$

where $\mathbf{H}_{ij}$ measures the cosine similarity between the i-th query patch and the j-th target patch, and $\mathbf{H} \in \mathbb{R}^{(H_qW_q)\times(H_tW_t)}$. For each query patch, its most similar target patch is determined by $j^* = \arg\max_j \mathbf{H}_{ij}$ and the corresponding 2D coordinates are recovered by mapping the linear indices of i and j^* back to their spatial patch locations in I_q and I_t, respectively.

To restrict correspondence to the object region, the query mask M_q is projected onto the DINOv3 patch grid to select foreground patches. Each valid patch i within M_q forms the point set $\mathcal{P}_q = \{\, i \mid (x_i, y_i) \in M_q \,\}$, and its best matched target point set $\mathcal{P}_t$ via feature matching. This foreground-constrained matching suppresses background noise and concentrates on the object region, enhancing geometric precision for subsequent stratification.

Point Stratification. Although applying M_q suppresses some noises, $\mathcal{P}_t$ may still be redundant or locally clustered. Thus, we further apply a stratified sampling process that enforces a minimum pairwise distance τ among $\mathcal{P}_q$ and $\mathcal{P}_t$:

$$\mathcal{P'}_t = \{\, p_i \mid \|p_i - p_j\|_2 > \tau, \forall j < i \,\}, \tag{2}$$

resulting in a compact yet spatially diverse set of high-quality correspondences $\mathcal{P'}_t$.

Coordinate Transformation and Prompt Encoding. The obtained sparse correspondences $\mathcal{P'}_t$ are defined in the original image coordinate system. To maintain consistency with SAM2's internal representation, they are linearly transformed to the canonical coordinate space using a deterministic geometric projection $\Pi(\cdot)$: $\widetilde{\mathcal{P}}_t = \Pi(\mathcal{P'}_t; (H_t^{\text{orig}}, W_t^{\text{orig}}))$. Finally, these coordinates $\widetilde{\mathcal{P}}_t$ are fed into the prompt encoder, resulting in the coordinate-based anchor prompt P^{q2t}_{anchor} for the target view.

3.3. Cross-View Visual Prompt Generator

In contrast to V^2-Anchor, which focuses on geometric correspondence, the V^2-Visual is designed to capture appearance-based cues for identifying visually similar objects across views. We use region-level representations to guide segmentation similar to text-guided SAM [47], and further introduce VPMatcher to bridge the cross-view appearance gap explicitly. Specifically, we first apply the SAM2 encoder $\phi(\cdot)$ to extract visual feature maps $\phi(I_q)$ and $\phi(I_t)$ from the query and target views, respectively. During training, we perform mask pooling on these feature maps using the corresponding object masks M_q and M_t to obtain region-level features:

$$\mathbf{v}_q = \text{MaskP}(\phi(I_q), M_q), \mathbf{v}_t = \text{MaskP}(\phi(I_t), M_t). \tag{3}$$

Visual Prompt Matcher (VPMatcher). The detailed illustration of VPMatcher is given in Fig. 3. It consists of two branches, a *Feature Mapping Branch* and a *Structural Mapping Branch*. The former learns semantically consistent representations across views, while the latter reconstructs object masks to maintain structural coherence [23].

(a) Feature Mapping Branch. Formally, given the fused prompt embedding $\mathbf{p}_f = \mathbf{v}_q + F_{\text{mask}}(M_q)$ obtained from the lightweight CNN-based mask encoder

$F_{\text{mask}}(\cdot)$, the feature mapping branch applies linear projections to form the query, key, and value spaces: $\mathbf{q}, \mathbf{k}, \mathbf{v} = W_q \mathbf{p}_f, W_k \phi(I_q), W_v \phi(I_q)$. The attention weights as: $\alpha = \text{Softmax}\left(\frac{\mathbf{q}\mathbf{k}^T}{\sqrt{D_e}}\right)$, where D_e is the embedding dimension. These weights are modulated by a spatial gating function to suppress background noise. The attention-weighted features are refined via multiple Transformer-based cross-attention encoder layers and a residual multi-layer perceptron, yielding the predicted cross-view embedding $\hat{\mathbf{v}}_c$.

(b) Structural Mapping Branch. In parallel, the structural mapping branch reconstructs the target mask from the coarse geometric prior [27, 45]. The input mask M_q is downsampled by $F_{\text{prior}}(\cdot)$ to a latent representation $\mathbf{m}_{\text{prior}}$. The prompt embedding $\mathbf{v}_q$ is projected to modulation parameters (γ, β) through a feature-wise linear modulation [3] layer $\Phi_{\text{cond}}(\cdot)$, injecting semantic conditioning into the mask space:

$$\tilde{\mathbf{m}} = \mathbf{m}_{\text{prior}} \odot (1 + \tanh(\gamma)) + \beta + F_{\text{mask}}(M_q). \quad (4)$$

Finally, the decoder $F_{\text{dec}}(\cdot)$ progressively upsamples $\tilde{\mathbf{m}}$ to generate the predicted cross-view mask $\hat{M}_c$. Furthermore, we can obtain new region features—i.e., the final visual prompt based on the predicted spatial mask as $\mathbf{v}_{c'} = \text{MaskP}(\phi(I_q), \hat{M}_c)$. This leads to the appearance-guided prompt, defined as $P^{q2t}_{visual} = \text{MLP}\left([\hat{\mathbf{v}}_c, \mathbf{v}_{c'}]\right)$. Here, the query and target prototypes are concatenated and projected through a lightweight MLP to form the final visual prompt.

3.4. Multi-Expert Training

Intuitively and empirically, we find that different prompts specialize in distinct forms of reasoning: V^2-Anchor is grounded in geometric structure, while V^2-Visual focuses on visual appearance. To leverage their complementary strengths and handle diverse cross-view scenarios, where both localization and visual association are crucial, especially in challenging cases, we design a multi-expert training mechanism that adaptively selects the most suitable expert under varying scene conditions.

To that end, as introduced in Sec. 3.1, three experts are constructed and trained, i.e., *Anchor Expert*, *Visual Expert*, and *Fusion Expert*. More specifically, all of them are using the same mask decoder architecture but different prompts as input to condition the mask decoder: Anchor Expert is using the geometry-aware prompt P^{q2t}_{anchor} generated from V^2-Anchor; Visual Expert is using visual-guided prompt P^{q2t}_{visual} predicted from V^2-Visual, and Fusion Expert takes both P^{q2t}_{anchor} and P^{q2t}_{visual}, and fuse them into a unified prompt embedding. For training, note that all steps in V^2-Anchor are non-learnable and directly output coordinate prompts P^{q2t}_{anchor} in the target view. Since these coordinate-based prompts are already supported by SAM2, we keep their corresponding expert entirely training-free by reusing the pretrained SAM2 mask decoder. In contrast, the Visual and Fusion Experts introduce the VPMatcher with new learnable parameters, and their resulting visual or visual–anchor prompts extend beyond the original SAM2 design. Therefore, we train the VPMatcher and mask decoder to better adapt to our task and datasets, while keeping the SAM2 encoder frozen. Both the Visual and Fusion Experts are optimized using the loss functions described below.

Loss Functions. Our training objective consists of three components: a visual contrastive loss $\mathcal{L}_v$, a structural constraint loss $\mathcal{L}_s$, and a mask prediction loss $\mathcal{L}_m$. The total loss $\mathcal{L}$ is defined as a weighted combination of these terms:

$$\mathcal{L} = \lambda_1 \mathcal{L}_v(\hat{\mathbf{v}}_c, \mathbf{v}_t) + \lambda_2 \mathcal{L}_s\left(\hat{M}_c, M_t\right) + \lambda_3 \mathcal{L}_m\left(\hat{M}_t, M_t\right), \quad (5)$$

where λ_1, λ_2, and λ_3 are the balancing coefficients.

(a) Visual Contrastive Loss. We employ contrastive loss across different views to enforce cross-view region-level features mapping in the same setting as [32, 33]. This loss encourages positive pairs to be close together and negative pairs to be far apart in the embedding space, thereby forming compact, well-separated feature clusters. Formally, it is defined as:

$$\mathcal{L}_v = -\frac{1}{N}\sum_{i=1}^{N}\left[\log \frac{\exp\left(\text{sim}(\mathbf{v}_c, \mathbf{v}_t)/\tau\right)}{\sum_{k=1}^{N}\exp\left(\text{sim}(\mathbf{v}_c, \mathbf{v}_t^k)/\tau\right)} + \log \frac{\exp\left(\text{sim}(\mathbf{v}_t, \mathbf{v}_c)/\tau\right)}{\sum_{k=1}^{N}\exp\left(\text{sim}(\mathbf{v}_t, \mathbf{v}_t^k)/\tau\right)}\right]. \quad (6)$$

where $\text{sim}(\cdot)$ denotes the cosine similarity function and τ is a temperature parameter.

(b) Mask Prediction Loss. Both the spatial constraint loss $\mathcal{L}_s$ and the mask loss $\mathcal{L}_m$ are designed based on mask-level supervision to ensure accurate spatial localization and structural consistency. The mask loss combines a pixel-wise cross-entropy term [26] and a region-level Dice loss [40]:

$$\mathcal{L}_m = \mathcal{L}_{CE}(\hat{M}, M) + \mathcal{L}_{Dice}(\hat{M}, M), \quad (7)$$

where $\mathcal{L}_{CE}$ penalizes pixel-wise classification errors, and $\mathcal{L}_{Dice}$ measures the overlap between the predicted and ground-truth masks.

(c) Structural Constraint Loss. Specifically, we also applied the mask loss for constraint cross-view structural mapping. $\mathcal{L}_s$ imposes a constraint on the cross-view structural mapping in VPMatcher, encouraging the model to learn a fixed spatial transformation.

3.5. Post-hoc Cyclic Consistency Selector

Our PCCS is designed to dynamically identify the most reliable expert, based on the bidirectional consistency between the query and target masks. PCCS projects each expert's

Method	Ego2Exo			Exo2Ego				Num. Param. (M)	
	IoU↑	Cont.A↑	Loc.E↓	IoU↑	Cont.A↑	Loc.E↓	Total-IoU↑	Total	Train
PSALM [52] (Zero-shot)	7.4	0.121	0.266	2.1	0.058	0.294	4.8	1587.1	0
CMX [50]	6.8	0.137	0.110	12.0	0.177	0.166	9.4	138.0	17.3
XSegTx [12]	18.9	0.386	0.070	27.1	0.358	0.104	23.0	12.1	3.6
XMem [12]	19.3	0.262	0.151	16.6	0.240	0.160	18.0	62.2	62.2
XMem + XSegTx [12]	34.9	0.559	0.038	25.0	0.237	0.117	30.0	75.6	67.1
Ref-SAM*	29.2	0.452	0.077	42.2	0.502	0.096	37.8	224.8	4.3
ObjectRelator [10]	35.3	0.540	0.036	40.3	0.500	**0.068**	37.8	1587.3	1587.3
O-MaMa (k-NN) [31]	31.9	0.414	0.195	30.9	0.373	0.127	31.4	154.0	0
O-MaMa [31]	42.6	0.590	**0.033**	44.1	0.524	0.082	43.4	165.6	11.6
♣ **V²-SAM** (Single-Expert)	44.5	0.607	0.055	47.3	0.552	0.092	45.9	531.3	7.6
♣ **V²-SAM** (Muti-Experts)	**46.3**	**0.616**	0.056	**49.6**	**0.575**	0.078	**48.0**	543.4	15.3

Table 1. Results on the Ego-Exo4D Correspondences v2 test split.

Method	$\mathcal{J}\&\mathcal{F}_m$ ↑	$\mathcal{J}_m$ ↑	$\mathcal{F}_m$ ↑
SiamMAE [14]	60.7	58.4	62.9
CrocoV2 + Cont. Pretrain [44]	40.0	37.4	42.5
Probabilistic Warp Consistency [43]	42.9	42.6	42.7
DINO ViTs/8 [4]	64.5	61.6	67.5
DINO ViTb/8 [4]	66.4	63.7	69.2
DINOv2 + Reg ViTb/14 [6]	62.1	59.6	64.8
PCC [1]	70.2	67.8	72.7
♣ **V²-SAM (Ours)**	**78.8**	**76.5**	**81.0**

Table 2. Comparison of video object correspondence on DAVIS-2017 Val with a temporal gap of 20 frames.

Setup	Method	Dataset	IoU↑
ZSL	XSegTx [12]	COCO / Self-Gen. Pairs	1.5
	SEEM [55]	COCO Panoptic, RefCOCO/+g	2.5
	PSALM [53]	COCO Panoptic, RefCOCO/+g, etc.	14.2
	PSALM [53]	Ego-Exo4D	39.9
	ObjectRelator [10]	Ego-Exo4D	42.8
	♣ **V²-SAM** (Single-Expert)	Ego-Exo4D	66.4
	♣ **V²-SAM** (Multi-Experts)	Ego-Exo4D	**77.2**

Table 3. Comparison of zero-shot (ZSL) object segmentation on HANDAL-X.

prediction $\hat{M}_{t_k}$ back to the query view via the V²-Anchor to assess geometric consistency with M_q. Formally, this inverse correspondence process can be expressed as:

$$P_k^{t2q} = \text{V}^2\text{-Anchor}\left(I_t, I_q; \hat{M}_{t_k}\right), \tag{8}$$

where $\hat{M}_{t_k}$ denotes the prediction from the k-th expert, and P_k^{t2q} represents the back-projected points on the query image corresponding to pixels within that prediction.

For cyclic consistency validation, we compute the average distance between the back-projected points and reference points sampled from the query mask as a proxy score to select the expert. Unlike prior cyclic schemes that reconstruct a query-view mask, our PCCS operates directly at the point level, avoiding an extra decoding pass while preserving selection accuracy. This lightweight validation enables effective expert selection during inference.

4. Experiments

4.1. Experimental Setup

Datasets and Evaluation Metrics. We conduct experiments on three representative benchmarks: Ego-Exo4D [12] for ego–exo object correspondence, DAVIS-17 [36] for video object tracking, and HANDAL-X [13] for robotic-ready cross-view segmentation. Ego-Exo4D is evaluated using *Mean Intersection over Union (mIoU)*, *Continuity Accuracy (Cont.A)*, and *Localization Error (Loc.E)*. For DAVIS-17, we report *Region Similarity* ($\mathcal{J}\uparrow$) and *Contour Accuracy* ($\mathcal{F}\uparrow$). HANDAL-X is evaluated with mIoU.

Implementation Details. We employ the SAM2-Hiera-Large model as the base segmentation backbone, and a DINOv3 ViT-L/16 Encoder as the feature extractor. All experiments were completed on eight NVIDIA H-series GPUs. The batch size for each GPU is set to 16, and with 4×10^{-5} learning rate. The loss weights are set to $\lambda_1 : \lambda_2 : \lambda_3 = 1 : 1 : 10$. However, during the first 4K training steps, we adopt a contrastive learning objective with $\lambda_1 = 100$ to enhance feature representation. During inference, all three experts operate in parallel, generating candidate masks conditioned on the visual and geometric anchor prompts. The final prediction is determined by our PCCS, with all evaluations performed using a batch size of 1.

4.2. Main Results

Baselines and Competitors. We compare V²-SAM with representative baselines on the three correspondence benchmark. XSegTx and XMem [12] are the official baselines that extend co-segmentation and temporal memory models for cross-view propagation. CMX [50] fuses multi-modal inputs through a transformer-based decoder. PSALM [52] integrates a large language model with Mask2Former [5]

Variant	Ego2Exo IoU↑	Exo2Ego IoU↑	Total IoU↑
Anchor Expert			
w/o V^2-Anchor	0.8	2.1	1.5
w/ V^2-Anchor (Ours)	**38.7**	**41.6**	**40.1**
Visual Expert			
w/o V^2-Visual	1.0	5.0	3.0
w/ V^2-Visual (Ours)	**36.2**	**46.6**	**41.4**

Table 4. Ablation on individual experts.

Method	Ego2Exo IoU↑	Exo2Ego IoU↑	Total IoU↑
A: Anchor Expert	38.7	41.6	40.1
B: Visual Expert	36.2	46.6	41.4
C: Fusion Expert	44.5	47.3	45.9
A + B	42.7	48.2	45.5
A + B + C	**46.3**	**49.6**	**48.0**

Table 5. Results of different expert decoder combinations on v2 test split. A: Anchor Expert, B: Visual Expert, C: Fusion Expert.

for zero-shot segmentation, while ObjectRelator [10] enhances PSALM with cross-view relational reasoning. Ref-SAM* is a SAM2-adapted baseline via visual reference-guided prompts. The recent O-MaMa [31] treats cross-view segmentation as a mask-matching task by generating FastSAM candidates and selecting the best match via contrastive learning. All methods are evaluated on the Ego-Exo4D Correspondences v2 test split.

Results on Ego-Exo4D. The comparison results are summarized in Tab. 1. We report IoU↑, Cont.A↑, and Loc.E↓ for both *Ego2Exo* and *Exo2Ego* directions, along with the averaged total IoU and parameter statistics. Among existing baselines, O-MaMa achieves the highest IoU of 42.6/44.1 on the two directions, while ObjectRelator attains 35.3/40.3 with substantially larger model size (1.6B parameters). Our proposed V^2-SAM (Multi-Experts) establishes a new state-of-the-art, achieving 46.31 IoU on Ego2Exo and 49.61 on Exo2Ego, surpassing O-MaMa by 3.7 and 5.5 IoU points on the Ego2Exo and Exo2Ego directions, respectively. Even the single-expert version already surpasses all prior baselines, achieving a total IoU of 45.9, which is 2.5 points higher than O-MaMa (43.4 IoU), demonstrating the strong capacity of our model. Compared to the Ref-SAM*, visual-reference guided segmentation struggles with cross-view segmentation, particularly in the Ego2Exo task, indicating that our framework addresses the shortcomings in visual guidance and achieves further improvements on this basis. Notably, our multi-expert model achieves this performance with only 543M total parameters and 15M trainable parameters, which is $\sim$1% of ObjectRelator's parameter count yet yields a +10.2 total IoU improvement. These results validate the effectiveness of our Multi-Prompt Experts.

Results on DAVIS-17. To further assess the generality of our framework for cross-view correspondence, we evaluate it on DAVIS-17, a standard benchmark for video object tracking. Following [1], we evaluate correspondence accuracy on all DAVIS-17 frame pairs separated by 20 frames, and consider only objects that are simultaneously visible in both views. We report performance using the standard $\mathcal{J}\&\mathcal{F}_m$, $\mathcal{J}_m$, and $\mathcal{F}_m$ metrics. As shown in Tab. 2, our V^2-SAM achieves the highest performance on DAVIS-17 across all evaluation metrics, surpassing all existing self-supervised correspondence models by a clear margin. V^2-SAM also achieves strong correspondence quality across large temporal gaps, highlighting its potential for robust object tracking under significant viewpoint shifts.

Results on HANDAL-X. HANDAL-X contains real-world, robotics-ready manipulable objects, offering a more practical testbed for Human2Robot transfer than Ego–Exo4D. Following [10], we evaluate V^2-SAM in the ZSL setting using a model trained on Ego–Exo4D. Tab. 3 shows that V^2-SAM demonstrates strong cross-view generalization, even the single-expert variant reaches an IoU of 66.4, substantially outperforming PSALM (39.9 IoU) and ObjectRelator (42.8 IoU). The multi-experts version further increases performance to 77.2 IoU.

4.3. Ablation Study

Ablation on Individual Expert. To evaluate the contribution of each expert, we compare it against its simplified baseline. 1) For the *Anchor Expert*, we replace V^2-Anchor with a simplified baseline that directly uses the centroid of M_q as the prompt for target-view prediction. For the *Visual Expert*, V^2-Visual aligns ego–exo object representations from both feature and structural perspectives. To evaluate its effectiveness, we remove the Visual Prompt Matcher and directly use the mask-pooled query-view feature as the prompt for cross-view segmentation. Tab. 4 summarizes the results. For Anchor Expert, removing V^2-Anchor leads to near-zero performance (1.5 total IoU), as the model entirely loses the ability to localize geometric anchors across views. For Visual Expert, removing the V^2-Visual also causes drastic degradation (3.0 total IoU), as directly using query-view features cannot provide meaningful cues for the target view. In contrast, our Visual Expert with V^2-Visual achieves strong performance (41.4 total IoU), effectively bridging the cross-view appearance gap.

Ablation on Experts Combinations. As shown in Tab. 5, due to their single-prompt nature, neither the Anchor Expert nor the Visual Expert achieves strong performance on Ego2Exo or Exo2Ego when evaluated individually. Integrating the two experts produces a notable performance gain (Ego2Exo: 38.7/36.2 $\rightarrow$ 42.7), confirming that anchor and visual cues provide complementary cross-view information.

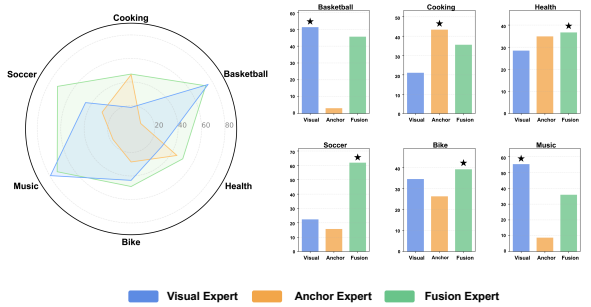


Figure 4. Comparison of Anchor, Visual, and Fusion Experts across different scenes. Left: per-scene IoU radar plot for the three experts. Right: per-scene Win% bars showing PCCS selections.

The Fusion Expert, which reasons jointly over both anchor and visual prompts, achieves balanced performance and robustly handles diverse scene conditions, resulting in performance that is on par with the two-expert combination. Finally, integrating all three experts within our Multi-Prompt Experts framework produces the best overall results across both Ego2Exo and Exo2Ego directions.

Expert Performance Analysis. To better understand how each expert behaves across different scene types, Fig. 4 visualizes their performance on the Ego2Exo task. Results are computed on samples where the IoU difference exceeds 0.3, highlighting cases with substantial disagreement and allowing clearer analysis of expert specialization. We observe that the 1) Anchor Expert demonstrates strong performance in structured and relatively static environments such as Cooking and Health. However, its effectiveness diminishes in scenes involving large motion or significant appearance change. In contrast, 2) the Visual Expert excels in dynamic and human-centric scenarios such as Basketball and Music, where appearance cues, semantic consistency, and deformation tolerance are crucial. 3) The Fusion Expert shows consistently broad coverage across all scene categories in the radar plot, indicating stable and balanced performance across diverse scenes.

4.4. Qualitative Results

To provide a more intuitive understanding of V^2-SAM, we present qualitative examples for both the Ego2Exo (Fig. 5) and Exo2Ego (Fig. 6) tasks. For each sample, we visualize the predictions from the *Anchor Expert* (AE), *Visual Expert* (VE), and *Fusion Expert* (FE), getting the final mask selected by PCCS. 1) *Ego2Exo:* The VE may incorrectly segment visually similar objects in cluttered backgrounds with multiple distractors, whereas the AE can accurately localize the correct target guided by geometric cues. Conversely, in highly dynamic scenes such as basketball, the VE remains reliable. 2) *Exo2Ego:* object manipulation scenes often involve substantial hand occlusion in the ego view requires



Figure 5. Ego2Exo qualitative results. From left to right: query view, predictions from the Anchor Expert, Visual Expert, and Fusion Expert, followed by the final output selected by the PCCS.

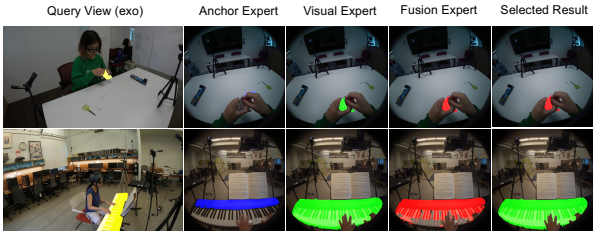


Figure 6. Exo2Ego qualitative results. From left to right: query view, predictions from the Anchor Expert, Visual Expert, and Fusion Expert, followed by the final output selected by the PCCS.

both geometric anchoring to locate the object and visual cues to determine its boundaries. The FE performs well under these conditions. For scenes where object appearance changes dramatically across views, such as music, the AE struggles to maintain consistent geometric correspondence, whereas the VE remains robust to such variations.

From the visualization results, we also observe that the Fusion Expert exhibits stable performance across diverse scene conditions, benefiting from its use of complementary prompts. PCCS further exploits cross-view mask consistency to reliably select the best prediction for each instance.

5. Conclusions

In this work, we presented V^2-SAM, a unified framework that extends the powerful segmentation foundation model SAM2 to the challenging task of cross-view object correspondence, with a particular focus on the Ego–Exo setting. To overcome the inherent difficulties posed by drastic viewpoint and appearance variations, we introduced two complementary prompt generators: V^2-Anchor, which restores spatially grounded prompting through geometry-aware correspondence, and V^2-Visual, which enhances appearance-guided understanding via a novel visual prompt matcher. Building on these, we designed a multi-prompt expert framework and a Post-hoc Cyclic Consistency Selector that adaptively selects the optimal expert for each instance. Extensive experiments on Ego-Exo4D, DAVIS-2017, and HANDAL-X benchmarks demonstrate that V^2-SAM achieves state-of-the-art performance across diverse cross-view correspondence tasks.

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